

Critical review: ICBB-AMRSS AMR surveillance dashboard

The project is a **promising, functioning surveillance prototype with a substantial feature base**, but it is not yet a decision-grade national One Health platform or a clinical decision-support system. The strongest next move is not adding more charts. It is making every record traceable, every metric defensible, every alert actionable, and every claim about clinical or national readiness verifiable.

What is already worth building on

The codebase has a coherent starting shape: separate sample and AST records, identifiers that link an isolate's test results back to a sample, source categories for human, animal, food, environment and aquaculture, and fields for location, dates, laboratory, MIC and disk-diffusion results. It also has distinct PPS, human antimicrobial-use (AMU) and animal-consumption (AMC) structures, a Kobo ingestion path, data-upload validation, WHONET export code, and a branded role-restricted Streamlit application. These are meaningful foundations, not just mock-up screens.

The local database snapshot I inspected is populated: it contained 1,040 sample rows and 4,588 AST rows, alongside PPS, AMU and AMC records. Treat these only as **local snapshot counts**; I cannot establish whether they are real surveillance observations, test/demo data, representative, complete or the current live production totals. The alert and prediction tables existed but had no rows in that local snapshot.

The public URL, gh-amr-dashboard.streamlit.app, initially displayed Streamlit's sleeping state and then loaded the branded sign-in page. With the credentials you supplied, I signed in and confirmed that the account exposes the application navigation, including Administration. Following your suggestion, I clicked Home and other tabs; Home and the AMU, AMC and Alerts views **did render**, so my earlier characterization of a persistent loading lock was incorrect. The first load was delayed, and during page changes the previous screen remained visible beneath a loading/analysis indicator until the next view finished rendering. I then signed out and confirmed that the login page returned.

At the time I inspected Home, it displayed 1,020 samples, 4,568 AST records, 26 organisms, 30.5% resistance, 0.0% MDR and 0 alerts. The sidebar freshness label read **“Updated: 28 Mar 2026, 15:45 • 182d ago”** ; unless this is intentionally a frozen demonstration dataset, that is a serious surveillance-freshness issue. The Alerts page, by contrast, displayed 2 critical and 111 high-priority alerts, plus additional medium/low counts and alert details. This may reflect different definitions (for example, saved alerts versus dynamically generated signals), but the discrepancy is not explained in the interface. Reconcile the

tile’ s meaning with alert generation and display the dataset/ “as of” date beside headline metrics so government users can judge whether information is current.

The decisive gap: traceability stops before the patient and genome

The present data model links `samples.sample_id` to `ast_results.sample_id` and associates results with an `isolate_id`. That is a useful start, but the inspected schema has **no patient pseudonym, encounter/episode, accession number, animal/herd/flock identifier, sequence-run entity, genomic-result entity, or chain-of-custody/event history** (`src/db.py`, `sample/AST` tables around lines 340–379). Nor is there a structured specimen type in that table. The current model can show that an AST result came from a recorded sample and isolate; it cannot yet follow the full clinical or One Health journey you described. Build the next version around stable, separately identified entities rather than a single wide spreadsheet:

- **Sampling event and specimen:** persistent IDs; collection date/time; collector and site; specimen/material type; source sector; sampling purpose and protocol; collection method; location/catchment; and receipt, rejection or storage details.
- **Subject or source:** a pseudonymous patient key and encounter/episode for human samples; host species and production unit/flock/herd for animal samples; and structured water-body, treatment-plant, catchment and upstream/downstream identifiers for water/environmental samples. Keep direct identifiers and re-identification keys outside routine surveillance views, with restricted access.
- **Isolate and phenotype:** isolate ID linked to its originating specimen; organism identification and method; each AST observation; raw MIC/zone values and units; AST platform/QC status; interpretive standard and **exact version/year**; and whether the result was measured, interpreted by a validated rule, or manually verified.
- **Genomic analysis:** sequencing sample/library/run IDs; secure read-file location and checksum; sequencing and contamination/QC metrics; assembler and pipeline versions; reference databases and versions; analysis date/operator; species confirmation; resistance genes/mutations, plasmid/typing results and cluster method; and reviewer status. Link each genomic result to the isolate—not just to a patient or a chart. Store large FASTQ/assembly files in governed object storage, not as database blobs.
- **Events and provenance:** an append-only record of collection, lab receipt, culture, identification, AST, sequencing, corrections, approvals, notifications and transfers, with timestamps, actor/site, source system, reason for changes and previous/new values.

Retain the original uploaded record so every displayed figure can be reproduced from a specific data and ruleset version.

This creates the path you want: **source/subject → sampling event → specimen → isolate → AST phenotype → sequencing analysis → interpreted signal → response and outcome**. It also supports multiple isolates from one specimen, multiple AST panels over time, and more than one sequencing analysis without overwriting history.

Use a shared national core for IDs, geography, timestamps, provenance and laboratory references, with **domain-specific data dictionaries and denominators** for human clinical, animal, food, water and environmental surveillance. A human clinical culture, farm sampling round and wastewater sample are not interchangeable observations. Integrate them for a defined question, time window and geography; do not simply pool their counts or resistance percentages and call the result One Health evidence. WHO GLASS itself distinguishes routine clinical AMR and consumption modules from focused surveillance and studies, while the 2026 Quadripartite guidance is explicitly intended to help countries design context-appropriate integrated AMR/AMU surveillance [1](#) [4](#) .

Highest-priority corrections before clinical or policy use

1. Remove the unprotected login diagnostic

The sign-in page accepts `?debug=1` before authentication and displays the total user count plus example email addresses (`app.py` , lines 783–795). Remove this path from deployed code or require an authenticated administrator and a protected server-side diagnostic channel. This is the most immediate security fix I found. Then test tenant and role isolation with separate lab, national and admin accounts, including exports, caches and error messages—not only the pages that hide datasets in the UI.

2. Do not present the current drug labels as prescribing advice

`analytics.py` generates “PREFERRED” , “GOOD” , “CAUTION” and “AVOID” labels from pooled susceptibility percentages with a five-test minimum (around lines 724–761). That is not patient-specific prescribing, and the threshold logic does not account for infection site, patient characteristics, specimen/source, local formulary, drug availability, breakpoint version, sampling bias or uncertainty. **Hide these labels or relabel them as exploratory surveillance summaries until a national clinical governance group validates the method.** Do not use them to direct an individual treatment decision.

A clinically usable cumulative antibiogram needs a governed period and population, verified diagnostic isolates, a defined deduplication rule, organism-specific reporting and transparent counts. CLSI’ s M39 guideline standardizes the recording, analysis and presentation of cumulative susceptibility data; its published summary describes first-

isolate deduplication and the conventional 30-isolate reporting threshold ³ ⁵. The current antibiogram code deduplicates repeated rows by `isolate_id + antibiotic` (`src/antibiogram.py`, lines 73–76). That is not the same as selecting the first isolate of a given organism per patient per analysis period. There is also no patient key in the schema to implement that rule. Fix the identifiers and analysis first, then validate the outputs with the national reference laboratory and clinicians before describing them as clinical antibiograms.

The interpretation engine contains manually embedded “CLSI 2025” and “EUCAST 2025” tables (`src/interpretation.py`, beginning around line 15). Before any automated interpretation is relied upon, verify every organism/drug/method breakpoint against the authoritative source; record its standard, edition, organism group and applicable host; test boundary cases and unsupported combinations; and make any derived S/I/R result visibly auditable. Unknown or unsupported combinations should remain unclassified rather than inherit a default.

3. Rebuild the alert and MDR logic as a reviewed workflow

The alert module uses fixed resistance thresholds and a minimum of ten test rows for rate alerts (`src/alerts.py`, lines 54–60 and 197–242). The code also uses a generic hard-coded antibiotic-class map for MDR detection. These can be useful development signals, but an alert needs an explicit denominator, deduplicated surveillance unit, minimum-data rule, baseline, source population, and a clear explanation of why it fired. Generic thresholds should not be mistaken for outbreak definitions or action thresholds approved by Ghanaian public-health authorities.

Create an operational alert lifecycle: **machine signal → laboratory verification → epidemiologist/clinician review → assigned response owner → action and due date → resolution/evidence → closure**. Record timestamps so programme managers can see time from specimen collection to confirmed result, alert review, notification and response. Keep the generating dataset, rule version, affected observations and reviewer attached to each event. Pilot the sensitivity and false-alarm burden against historical data before any automatic notification or escalation.

4. Correct denominator handling in AMU, AMC and PPS

The AMU page plots raw quantities for regional comparisons and averages row-level `ddd_per_1000` values by period (`src/page_amu.py`, lines 64–105). Simple means across records can be misleading when sites have different patient-day denominators or reporting volumes. Store numerator, denominator, units, conversion factor and calculation method, then calculate a pooled rate from the appropriate totals. Show raw use alongside a clearly defined normalized rate. Do not compare regions using an unnormalized quantity alone.

The animal AMC page also averages row-level `mg_per_kg_biomass` by species (`src/page_amc.py` , lines 92–106). Verify that the underlying unit is active ingredient mass and calculate a correctly weighted measure from total active ingredient and the specified biomass denominator. Keep species, production type, route, purpose, reporting period and antimicrobial classification explicit. WHO GLASS provides standardized approaches for national AMC reporting, while hospital AMU and national sales/consumption are distinct measurement contexts ¹ .

There is a specific PPS upload risk: when a workbook contains multiple surveys, the upload code partitions prescription rows into roughly equal chunks by row order (`app.py` , lines 1993–2020). Unless every prescription is already grouped in exactly that order and evenly distributed, records can be assigned to the wrong survey. Require a `survey_id` foreign key on every prescription row and reject orphaned or ambiguous rows. Add a multi-survey test fixture before using combined PPS files.

5. Treat interoperability labels as unverified until tested

A WHONET export helper exists, but code maps unknown organism and antibiotic names to generated fallback codes (`src/whonet.py` , around lines 159–192). An export file existing does not by itself prove WHONET import compatibility or GLASS submission readiness. Version the export schema, use governed code sets, report unmapped values as errors rather than inventing codes, and test round-trips against reference files with human, veterinary and non-clinical specimen examples. WHO GLASS describes standardized, quality-oriented national data collection and sharing—not merely a file labelled “GLASS” ¹ .

What would make this genuinely useful to government

Government users need more than a national resistance heatmap. The system should answer: **where is the evidence reliable, what changed, how serious is it, who must act, and did the action work?** Make data-system performance a first-class surveillance product. Report site and sector coverage; expected versus received submissions; timeliness; completeness of core fields; duplicate, invalid and unlinked record rates; AST quality-control performance; sequence linkage; and missing denominators. Show reporting volume and uncertainty beside every rate, and allow users to distinguish “no signal” from “no data.”

For clinical decision-making, generate facility-specific, period-specific antibiograms that follow the accepted method and are approved by the responsible microbiology/clinical governance group. Link those summaries to antimicrobial stewardship and infection-prevention review, not direct automatic prescriptions. For animal health and food safety, track use by species, production class, indication and route alongside structured pathogen and food-chain sampling. For water and environment, capture sample matrix, source,

catchment, treatment stage, volume/method, season and relevant contextual measurements. These sector outputs should remain analytically distinct unless a documented integrated question justifies combining them.

Replace static “risk score” rankings with transparent indicators whose components can be inspected. Useful government views include burden and trend by defined population; geographic and sector coverage; reporting timeliness; confidence/precision; emerging or unusual resistance confirmed by reference testing; antimicrobial use/consumption with the correct denominator; and unresolved response actions. Track policy action against the current national action plan. The WHO library page lists a Ghana national action plan entry but gives no plan year in its visible title, so confirm the current approved Ghana plan and responsible indicator owners directly with the national AMR coordination structure before hard-coding targets ².

The key differentiator can be **linked, auditable One Health evidence**, not a claim that the system predicts transmission between people, animals, food and water. A shared organism or resistance gene across sectors is a signal for investigation; it is not by itself proof of a transmission route or source. If cross-sector genomic comparisons become a goal, agree sampling frames, reference-lab confirmation, genomic methods, thresholds, privacy rules and investigation procedures before presenting clusters as public-health events.

A practical build sequence

1. **Stabilize trust and safety first.** Remove the login debug leak and the unvalidated prescribing labels. Freeze a metric dictionary covering units, numerator, denominator, time period, deduplication, standard/version, suppression and intended use. Reconcile the exhibition walkthrough with what is actually implemented: label each feature as live, pilot, planned or illustrative. Add automated tests for the core validation, calculations, data access and export.
2. **Design the traceability model with users, not in another spreadsheet.** Convene human laboratories, reference laboratory, clinicians, infection prevention, veterinary/food and water/environmental teams, the national AMR coordinating body, data-protection/legal support and programme managers. Agree the minimum national event and identifier model, role-based access, retention, correction, data-sharing and alert authority. Confirm which current Ghana policy indicators the platform is meant to serve.
3. **Pilot the full chain on real workflows.** Choose a small, representative human clinical network plus one veterinary/food pathway and one water/environment pathway. Barcode or otherwise uniquely identify specimens and isolates; link phenotype and selected sequencing results; test offline/manual capture and upload; audit each handoff; and compare displayed indicators with a reference analyst’s independent

calculation. Evaluate missingness, delays, duplication, false alerts and staff workload—not only whether the screens render.

4. **Add the government action layer.** Build a national overview of data coverage and quality, validated trends, reviewed signals, AMU/AMC measures and assigned response actions. Use user-specific views so clinicians, laboratorians, field officers and policy leaders see the decisions they are responsible for. Provide a record of decisions and follow-up, not just charts and downloadable reports.
5. **Scale only after evidence and operating ownership exist.** Establish reference-laboratory support, external quality assessment, training, a funded maintenance and sequencing plan, service-level expectations, disaster recovery, security monitoring and an accountable national data steward. Expand by readiness and data quality, not a headline number of sites.

To make this a credible international product, keep country-specific lab lists, administrative geography, thresholds and reporting definitions configurable. Preserve a common traceability and provenance core, and publish a versioned, documented data contract and conformance test suite. Ghana can be the reference implementation; the claim of global portability should follow successful implementation and evaluation in more than one governance and laboratory context.

Bottom line

The project already has enough functional breadth to justify a serious pilot. Its next leap depends on **scientific validity, specimen-to-isolate-to-genome traceability, secure data governance, interoperable data and a visible public-health response loop**. Until those are in place, present it as a promising surveillance prototype—not a real-time national early-warning, patient-prescribing or genome-linked One Health system. That distinction will make the project more credible to Ghanaian decision-makers and international partners, not smaller.

References

- [1] WHO Global Antimicrobial Resistance and Use Surveillance System (GLASS)
- [2] WHO Library of national action plans
- [3] CLSI publishes new guideline CLSI M39—Analysis and Presentation of Cumulative Antimicrobial Susceptibility Test Data
- [4] Quadripartite guidance on One Health integrated surveillance of antimicrobial resistance and use
- [5] What’ s New in Antibigrams? Updating CLSI M39 Guidance with the Arrival of the 5th Edition